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Introduction to BMW

Model: All

Production: All

OBJECTIVES

After completion of this module you will be able to:

- Identify BMW models by their internal and external designations.
- Be able to decipher a VIN.
- Identify BMW engines and transmissions.
- Understand the BMW Group Numbering System.
- Recognize acronyms utilized internally by BMW.

BMW Model Identification

BMW uses two different methods of identifying its products:

- External Designations are for customers to identify BMW products. These are marketing designations commonly referred to as trunk badges.
- Internal Designations are used by BMW and center personnel to identify a particular production version.

External Designations

External designations located on the trunk or tailgate are the model names. Customers and the public identify BMW products by these names.



The trunk badges consist of a combination of numbers and letters. The badge shown is typical of 1, 2, 3, 4, 5, 6, 7 and 8 series vehicles. The first digit represents the series identification. The last two digits usually mark the engine displacement and/or engine performance level. The “e” represents eDrive or electric mobility.

With the introduction of the i3 and i8 vehicles in 2014, the letter “i”, is also used to represent electric drive (when used as the first letter).



The external designations are created by the sales and marketing department to position each model in a certain target segment of the vehicle sales market.

Occasionally there are examples where the actual engine displacement does not match the designation. This is a marketing decision to continue familiar name recognition from one version to the next and to create a spacing between models.

Examples are:

- F16 X6 xDrive50i - 4.4 liter
- F30 340i - 3.0 liter
- G12 750i/Li - 4.4 liter



There is another series of badges that begin with a letter followed by a single digit. These are found on:

- Roadsters “Z” (E52, E85, E89)
- SAV “X” (E53, E70, E83, F15)
- SAC “X” (E71, E72, F16, F26)
- Motorsport “M”
- Electric Drive “i” (i01, i12)



Additional letters sometime follow the badge numbers and are used to further identify each model. Examples include:

- xDrive - all-wheel drive
- L - long wheel base
- sDrive -sport/standard drive
- is - sport
- i - (stand alone) iPerformance
- i - (after numbers) Fuel Injected
- M - Motorsport
- ActiveHybrid
- GT - Gran Turismo
- e/eDrive - Electric Machine
- d - Diesel
- t - touring/sports wagon
- C or c - Coupe or Cabrio

Examples of these badges are:



Internal Designations

Internal designations are created and used by the Engineering and Technical Development departments to identify a new model during the design of the car. The use of the internal designations carries over to the Technical Support groups after production.

Internal designations refer to the entire vehicle series of a particular body design. For example, the E90 designation describes all 3 series cars of that design. BMW products start with an “E” (as of 2009 we started using the letter “F” on our vehicles) and are followed by a two digit number. Not all design concepts make production, so internal designations are usually not sequential. The table on the next page lists the internal designations, series number and the years produced.

The E designations can include an additional number separated by a slash (/) to indicate a different model within a particular chassis group. Example:

- E36/7- Z3
- E46/3- Sport Wagon
- E46/16- All-Wheel Drive
- E36/5- 318ti
- E46/2- Coupe

The slash was no longer used after the E46 models. Every model, after the E46, had their own designation. Example:

- E90 - 3 series sedan
- E91 - 3 series Sport Wagon
- E92- 3 series coupe
- E93 - 3 Series Convertible

The I designations came out in model year 2014, for vehicles with electric drive. BMW i represents future-oriented vehicles and mobility services whose premium character is consistently defined using sustainability. The BMW i3 (internal designation I01) is the first series model of the new brand, allowing emissions-free mobility in a premium car.

Engineering Designation	Series	Years Produced
E12	5	1975 - 1981
E21	3	1977 - 1983
E23	7	1978 - 1987
E24	6	1977 - 1989
E28	5	1982 - 1988
E30	3	1984 - 1993
E31	8	1991 - 1997
E32	7	1988 - 1994
E34	5	1989 - 1995
E36	3	1992 - 1999
E36/7	Z3 Roadster	1996 - 2002
E38	7	1995 - 2001
E39	5	1997 - 2003
E46	3	1999 - 2006
E52	8	2001 - 2003
E53	5 SAV (X5)	2000 - 2006
E60/61	5 Sedan/Wagon	2004 - 2010
E63/64	6 Coupe/Convertible	2004 - 2010
E65/66	7 Sedan/Long Wheel Base	2002 - 2008
E70	5 SAV (X5)	2007 - 2013
E71	6 SAC (X6)	2008 - 2014
E72	6 SAC ActiveHybrid	2010 - 2011
E82	1 Coupe	2008 - 2013
E83	3 SAV (X3)	2004 - 2010
E84	1 SAV (X1)	2013 - 2015
E85/86	Z4 Roadster/Coupe	2003 - 2008
E88	1 Convertible	2008 - 2013
E89	Z4 Roadster	2009 - 2016
E90/91	3 Sedan/Wagon	2006 - 2012
E92/93	3 Coupe/Convertible	2007 - 2013
F01/F02	7 Sedan/Long Wheel Base	2009 - 2015
F04	7 ActiveHybrid	2011 - 2012
F06	6 Gran Coupe	2013 -
F07	5 Gran Turismo	2010 - 2017
F10	5 Sedan	2011 - 2017
F12	6 Convertible	2011 - 2018

Engineering Designation	Series	Years Produced
F13	6 Coupe	2012 - 2017
F15	5 SAV (X5)	2014 - 2018
F16	6 SAC (X6)	2015 -
F22	2 Coupe	2014 -
F23	2 Convertible	2015 -
F25	3 SAV (X3)	2011 - 2017
F26	4 SAC (X4)	2015 - 2018
F30	3 Sedan	2012 - 2018
F31	3 Wagon	2014 -
F32	4 Coupe	2014 -
F33	4 Convertible	2014 -
F34	3 Gran Turismo	2014 -
F36	4 Gran Coupe	2015 -
F39	2 SAC (X2)	2018
F48	1 SAV (X1)	2016 -
F80	M3 Sedan	2015 -
F82	M4 Coupe	2015 -
F83	M4 Convertible	2015 -
F85	X5 M	2015 - 2018
F86	X6 M	2015 -
F87	M2 Coupe	2016 -
F90	M5 Sedan	2018 -
G01	3 SAV (X3)	2018 -
G02	4 SAC (X4)	2018 -
G05	5 SAV (X5)	2018 -
G07	7 SAV (X7)	2019 -
G12	7 Sedan	2016 -
G14	8 Convertible	2019 -
G15	8 Coupe	2018 -
G20	3 Sedan	2019 -
G29	Z4 Roadster	2019 -
G30	5 Sedan	2018 -
G32	6 Gran Turismo	2018 -
I01	i3	2014 -
I01s	i3s	2018 -
I12	i8 Coupe	2014 -

NOTES

PAGE

1 Series



E82

2008 - 2013

128i 2008 - 2013

135i 2008 - 2013



E88

2008 - 2013

128i 2008 - 2013

135i 2008 - 2013

2 Series



F22

2014 -	
228i	2014 - 2016
M235i	2014 - 2016
228xi	2015 - 2016
M235xi	2015 - 2016
230i/xi	2017 -
M240i/xi	2017 -



F23

2015 -	
228i	2015 - 2016
M235i	2015 - 2016
228xi	2015 - 2016
M235xi	2015 - 2016
230i/xi	2017 -
M240i	2017 -

3 Series



E21

1977 - 1983

320i 1977 - 1983



E30

1984 - 1993

318i 1984 - 1985
1991 - 1992

325e 1985 - 1988

325i 1987 - 1993



E36

1992 - 1999

318i 1992 - 1996

325i 1992 - 1995

323i 1998 - 1999

328i 1996 - 1999

	E46 1999 - 2006 <table> <tr> <td>323i</td><td>1999-2000</td></tr> <tr> <td>325i</td><td>2001-2006</td></tr> <tr> <td>328i</td><td>1999-2001</td></tr> <tr> <td>330i</td><td>2001- 2006</td></tr> <tr> <td>M3</td><td>2002 - 2006</td></tr> </table>	323i	1999-2000	325i	2001-2006	328i	1999-2001	330i	2001- 2006	M3	2002 - 2006						
323i	1999-2000																
325i	2001-2006																
328i	1999-2001																
330i	2001- 2006																
M3	2002 - 2006																
	E90 2006 - 2011 <table> <tr> <td>325i</td><td>2006 - 2007</td></tr> <tr> <td>330i</td><td>2006 - 2007</td></tr> <tr> <td>325xi</td><td>2006 - 2007</td></tr> <tr> <td>330xi</td><td>2006 - 2007</td></tr> <tr> <td>328i</td><td>2007 - 2011</td></tr> <tr> <td>335i</td><td>2007 - 2011</td></tr> <tr> <td>328xi</td><td>2007 - 2011</td></tr> <tr> <td>335xi</td><td>2007 - 2011</td></tr> </table>	325i	2006 - 2007	330i	2006 - 2007	325xi	2006 - 2007	330xi	2006 - 2007	328i	2007 - 2011	335i	2007 - 2011	328xi	2007 - 2011	335xi	2007 - 2011
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330i	2006 - 2007																
325xi	2006 - 2007																
330xi	2006 - 2007																
328i	2007 - 2011																
335i	2007 - 2011																
328xi	2007 - 2011																
335xi	2007 - 2011																
	E92 2007 - 2013 <table> <tr> <td>328i</td><td>2007 - 2013</td></tr> <tr> <td>335i</td><td>2007 - 2013</td></tr> <tr> <td>328xi</td><td>2007 - 2013</td></tr> <tr> <td>335xi</td><td>2007 - 2013</td></tr> <tr> <td>335is</td><td>2011 - 2013</td></tr> </table>	328i	2007 - 2013	335i	2007 - 2013	328xi	2007 - 2013	335xi	2007 - 2013	335is	2011 - 2013						
328i	2007 - 2013																
335i	2007 - 2013																
328xi	2007 - 2013																
335xi	2007 - 2013																
335is	2011 - 2013																

	E93 2007 - 2013 328i 2007 - 2013 335i 2007 - 2013 335is 2011 - 2013
	F30 2012 - 2018 328i 2012 - 2016 335i 2012 - 2015 320i/xi 2013 - 2018 328xi 2013 - 2016 335xi 2013 - 2015 ActiveHybrid 3 2013 - 2014 328d 2014 - 2018 328xd 2014 - 2018 340i 2016 - 2018 340xi 2016 - 2018 330e 2016 - 2018 330i/xi 2017 - 2018



G20

2019 -	
330i	2019 -
330i xDrive	2019 -
M340i	2019 -
M340i xDrive	2019 -

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3 Series Gran Turismo

	F34 2014 - 328xi 2014 - 2016 335xi 2014 - 2016 330xi 2017 - 340xi 2017 -

4 Series



F32

2014 -	
428i	2014 - 2016
428xi	2014 - 2016
435i	2014 - 2016
435xi	2014 - 2016
430i/xi	2017 -
440i/xi	2017 -



F33

2014 -	
428i	2014 - 2016
428xi	2014 - 2016
435i	2014 - 2016
435xi	2015 - 2016
430i/xi	2017 -
440i/xi	2017 -



F36

2015 -	
428i	2015 - 2016
428xi	2015 - 2016
435i	2015 - 2016
435xi	2015 - 2016
430i/xi	2017 -
440i/xi	2017 -

5 Series



E12

1975 - 1981

530i	1975 - 1978
528i	1979 - 1981



E28

1982 - 1988

528e	1982 - 1988
533i	1983 - 1984
535i	1985 - 1988
524td	1985 - 1986



E34

1989 - 1995

525i	1989-1995
530i	1994-1995
540i	1994-1995



E39

1997 - 2003

525i	2001
528i	1997 - 2000
530i	2001 - 2003
540i	1997 - 2003



E60

2004 - 2010

525i	2004 - 2007
530i	2004 - 2007
545i	2004 - 2007
550i	2006 - 2010
525xi	2006 - 2007
530xi	2006 - 2007
528i	2008 - 2010
535i	2008 - 2010
535xi	2008 - 2010



F10

2011 - 2016

528i / xi	2011 - 2016
535i / xi	2011 - 2016
550i / xi	2011 - 2016
535i Hybrid	2012 - 2013
535d / xDrive	2014 - 2016
ActiveHybrid 5	2014 - 2016



G30

2017 -

530i / xDrive 2017 -

540i / xDrive 2017 -

530e / xDrive 2018 -

540d xDrive 2018 - 2019

M550i xDrive 2018 -

5 Series Gran Turismo



F07

2012 - 2017

535i 2012 - 2017

550i 2012 - 2015

535xi 2013 - 2017

550xi 2013 - 2017

6 Series



E24

1977 - 1989

630CSi	1977
633CSi	1978 - 1984
635CSi	1985 - 1989
L6	1987



E63

2004 - 2010

645Ci	2004 - 2005
650i	2006 - 2010



E64

2004 - 2010

645Cic	2004 - 2005
650i	2006 - 2010



F12

2011 - 2018

650i	2011 - 2018
640i	2012 - 2018
650xi	2012 - 2018
640xi	2014 - 2018



F13

2012 - 2017

640i	2012 - 2017
650i	2012 - 2017
650xi	2012 - 2017
640xi	2014 - 2017



F06

2013 -

640i	2013 -
640xi	2014 -
650i	2013 -
650xi	2013 -

	F06 2015 - Alpina B6 2015 -

7 Series



E23

1978 - 1987

733i 1978 - 1984

735i 1985 - 1987

L7 1986 - 1987



E32

1988 - 1994

735i 1988-1991

740i 1992-1994

750iL 1988-1994



E38

1995 - 2001

740i 1995-2001

750iL 1995-2001



E65

2002 - 2008

745i 2002 - 2005

750i 2006 - 2008

760i 2005 - 2008



E66

2002 - 2008

745Li 2002 - 2005

750Li 2006 - 2008

760Li 2003 - 2008



F01/F02

2009 - 2015

740i/Li 2011 - 2015

750i/Li 2009 - 2015

750i/Li xDrive 2010 - 2015

760Li 2010 - 2015

740Lxi 2013 - 2015

740Li Hybrid 2013 - 2015

ActiveHybrid 7 2014 - 2015

740Lxd 2015 - 2015



F01/F02

2011 - 2015

BMW ALPINA B7

2011 - 2015

BMW ALPINA B7 xDrive

2011 - 2015

BMW ALPINA B7 LWB

2011 - 2015

BMW ALPINA B7 LWB xDrive

2011 - 2015



F04

2011 - 2013

ActiveHybrid 750i

2011 - 2013

ActiveHybrid 750Li

2011 - 2013



G12

2016 -

750i 2016 -

750xi 2016 -

740i 2016 -

740xi 2017 -

740e 2017 -

M760i 2017 -

8 Series



E31

1991 - 1997

840Ci 1994 - 1997

850i 1991 - 1994

850Ci 1995 - 1997

850CSi 1994 - 1995



G14

2019 -

M850 xDrive 2018 -



G15

2018 -

M850 xDrive 2018 -

Touring - Sports Wagon



E36/5

1995 - 1999

318 ti 1995 - 1999



E46

2000 - 2005

323iT 2000

325iT 2001 - 2005

325xiT 2001 - 2005



E91

2006 - 2012

325xiT 2006

328xiT 2007 - 2012

328iT 2007 - 2012



F31

2014 -

328xi

328xd

330xi

2014 - 2016

2014 - 2018

2017 -



E34

1992 - 1995

525iT 1992 - 1995

530iT 1994 - 1995



E39

1999 - 2003

528iT 1999 - 2000

525iT 2001 - 2003

540iT 1999 - 2003



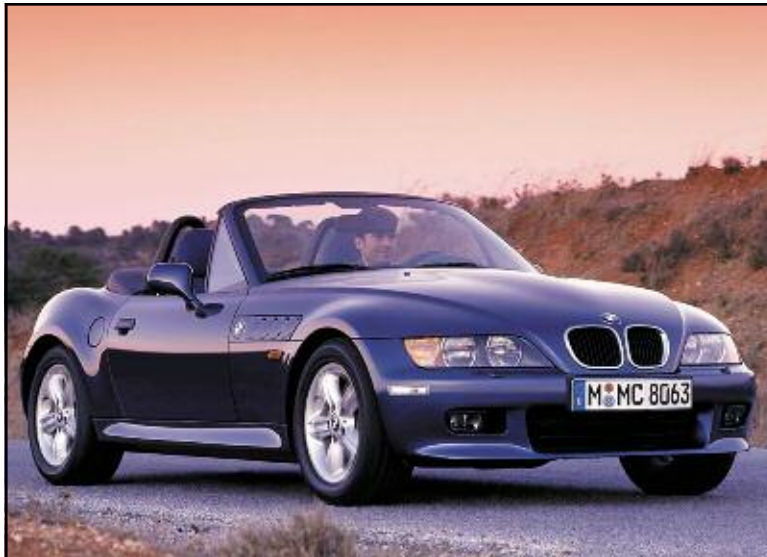
E61

2006 - 2010

530xiT 2006 - 2007

535xiT 2008 - 2010

Z Series



E36/7 Z3

1996 - 2002

Z3 1.9	1996 - 1998
Z3 2.3	1999 - 2002
Z3 2.8	1997 - 2000
Z3 3.0	2001 - 2002
Z3 Coupe 2.8	1999 - 2000
Z3 Coupe 3.0	2001 - 2002



E85/E86 Z4

2003 - 2008

Z4 2.5	2003 - 2005
Z4 3.0	2003 - 2005
Z4 3.0i	2006 - 2008
Z4 3.0si	2006 - 2008
Z4 Coupe	2006 - 2008



E89 Z4

2009 - 2016

Z4 sDrive28i	2009 - 2016
Z4 sDrive35i	2009 - 2016
Z4 sDrive35is	2011 - 2016



E52 Z8

2001 - 2003

Z8

2001 - 2003

ALPINA Z8

2003



G29 Z4

2019 -

Z4 sDrive30i

2019 -

Z4 M40i

2019 -

X Series



E84 X1

2013 - 2015

X1 sDrive28i 2013 - 2015

X1 xDrive28i 2013 - 2015

X1 xDrive35i 2013 - 2015



F48 X1

2016 -

X1 xDrive28i 2016 -

X1 sDrive28i 2017 -



F39 X2

2017 -

X2 sDrive28i 2018 -

X2 xDrive28i 2017 -



E83 X3

2004 - 2010

X3 2.5i	2004 - 2006
X3 3.0i	2004 - 2006
X3 3.0si	2007 - 2008
X3 xDrive30i	2009 - 2010



F25 X3

2011 -

X3 xDrive28i	2011 - 2017
X3 xDrive35i	2011 - 2017
X3 xDrive28d	2015 - 2016
X3 sDrive28i	2015 - 2017



G01 X3

2017 -

X3 xDrive30i	2017 -
X3 M40i	2017 -



F26 X4

2015 - 2018

X4 xDrive28i 2015 - 2018

X4 xDrive35i 2015 - 2016

X4 xDriveM40i 2016 - 2018



G02 X4

2018 -

X4 xDrive30i 2018 -

X4 M40i 2018 -



E53 X5

2000 - 2006

X5 4.4i 2000 - 2006

X5 3.0i 2001 - 2006

X5 4.6is 2002 - 2003

X5 4.8is 2004 - 2006



E70 X5

2007 - 2013

X5 3.0si	2007 - 2008
X5 4.8i	2007 - 2008
X5 xDrive30i	2009 - 2010
X5 xDrive35d	2009 - 2013
X5 xDrive35i	2011 - 2013
X5 xDrive50i	2011 - 2013



F15 X5

2014 -

X5 sDrive35i	2014 - 2018
X5 xDrive35d	2014 - 2018
X5 xDrive35i	2014 - 2018
X5 xDrive50i	2014 - 2018
X5 xDrive40e	2016 - 2018



G05 X5

2018 -

X5 xDrive40i	2018 -
X5 xDrive50i	2018 -



E71 X6

2008 - 2014

X6 xDrive35i 2008 - 2014

X6 xDrive50i 2008 - 2014



E72 X6

2010 - 2011

X6 xDrive50i Hybrid



F16 X6

2015 -

X6 sDrive35i 2015 -

X6 xDrive35i 2015 -

X6 xDrive50i 2015 -

	G06 X6
	G07 X7 2019 - X7 xDrive40i 2019 - X7 xDrive50i 2019 -

Motorsport Series



E26 M1

1976 - 1981



E82 M

2011



F87 M2

2016 -



E30 M3

1988 - 1992



E36 M3

1995 - 1999



E46 M3

2002 - 2006



E9x M3

2008 - 2013

E90 2008 - 2011

E92 2008 - 2013

E93 2008 - 2013



F80 M3

2015 - 2018

M3 CS 2018



F82 M4

2015 -

M4 CS 2018 -



F83 M4

2015 -



F82 M4 GTS

2016



E28 M5

1988



E34 M5

1991 - 1993



E39 M5

2000 - 2003



E60 M5

2006 - 2010



F10 M5

2013 - 2016



F90 M5

2018 -



E24 M6

1987 - 1988



E63/64 M6

2006 - 2010

E63 2006 - 2010

E64 2007 - 2010



F12 M6

2012 -



F13 M6

2013 - 2017



F06 M6 Gran Coupe

2014 -



E36/7 M Roadster

1998 - 2002



E36/7 M Coupe

1999 - 2002



E85 M Roadster

2006 - 2008



E86 M Coupe

2006 - 2008



E70 X5 M

2010 - 2013



E71 X6 M

2010 - 2014



F85 X5 M

2015 - 2018



F86 X6 M

2015 -

BMW I



I01 i3

2014 -	
i3 BEV	2014 -
i3 Rex	2014 -
i3s BEV	2018 -
i3s Rex	2018 -



I12 i8

2014 -	
i8 Coupe	2014 -



I15 i8

2018 -	
i8 Roadster	2018 -

VIN

VIN Location on Vehicle

The VIN is located on the vehicle in the following locations:

- Left lower corner of the instrument panel. This is viewed through the windshield from outside the vehicle.
- B pillar compliance label. On the drivers side door jamb.

Additionally the VIN is stamped into the chassis in one of two locations:

- Under the hood on right side of engine compartment bulkhead.
- Under the hood on right side shock tower.



Vehicle Production Dates



The production date of the vehicle is included in the B pillar compliance label.

Production date information is required quite often when a repair procedure affects only a certain range of vehicles. The range is based on production dates.

The date shown on the label reflects the actual month and year the vehicle was produced. It is not a model year indicator.

Vehicle Production Anti-Theft Labels

Since 1987, the National Highway Traffic Safety Administration (NHTSA) requires that the VIN be marked on specific parts of the car during manufacture for theft identification. Some models are exempt if the total number of vehicles imported is below a certain number.

If equipped, the label is also a reference point for the VIN. The label is placed on the following locations:

- Engine
- Hood
- Front bumper
- Front Fenders
- Quarter panels.
- Firewall
- Transmission
- Trunk lid
- Rear Bumper
- Doors
- Shock Tower



* Please reference SI B 00 21 86 (1267) dated June 1990 for more information on Anti-theft labels.

Vehicle Identification Numbers

Vehicle Identification Numbers are referred to as the VIN. BMW utilizes a VIN with a 17 character structure. The characters are grouped to include important vehicle information. Please take a look at the following simplified explanation:

WBA BR33 4 4 Y E A82019

Digit 1-3 are the manufacturer's code: ←

WBA = BMW AG, Munich. Germany

WBS = BMW M GmbH, Munich. Germany

4US = BMW Manufact. Corp. Spartanburg, SC. USA

5UX = BMW Manufact. Corp. Spartanburg, SC. USA

5UM = BMW Manufact. Corp. Spartanburg, SC. USA

Digits 4-7 are the model code: ←

Digit 8 is the restraint system identifier: ←

0 = Seat belts

1 = Seat belts and Driver airbag

2 = Seat belts and Dual airbags

3 = Seat belts, Second generation dual front airbags

4 = Seat belts, Dual-stage advanced front airbags

5 = Seat belts and Airbags (the number and type of airbags is vehicle dependent)

Digit 9 is a check digit determined by VIN: ←

Digit 10 is the model year identifier (started in 1980): ←

A = 1980 model year, sequentially progresses every year. The letters I, O, Q, U and Z are not used because they can be mistaken for numbers.

G=1986

N=1992

W=1998

4=2004

A=2010

G=2016

H=1987

P=1993

X=1999

5=2005

B=2011

H=2017

J=1988

R=1994

Y=2000

6=2006

C=2012

J=2018

K=1989

S=1995

1=2001

7=2007

D=2013

K=2019

L=1990

T=1996

2=2002

8=2008

E=2014

L=2020

M=1991

V=1997

3=2003

9=2009

F=2015

M=2021

Digit 11 is the Plant code: ←

A, F, K = (Plant 1.1) Munich Germany

C, B, D, G = (Plant 2.4) Dingolfing Germany

E, J, P = (Plant 6.1) Regensburg Germany

L, M = (Plant 10) Spartanburg USA

N = (Plant 9) Rosslyn South Africa

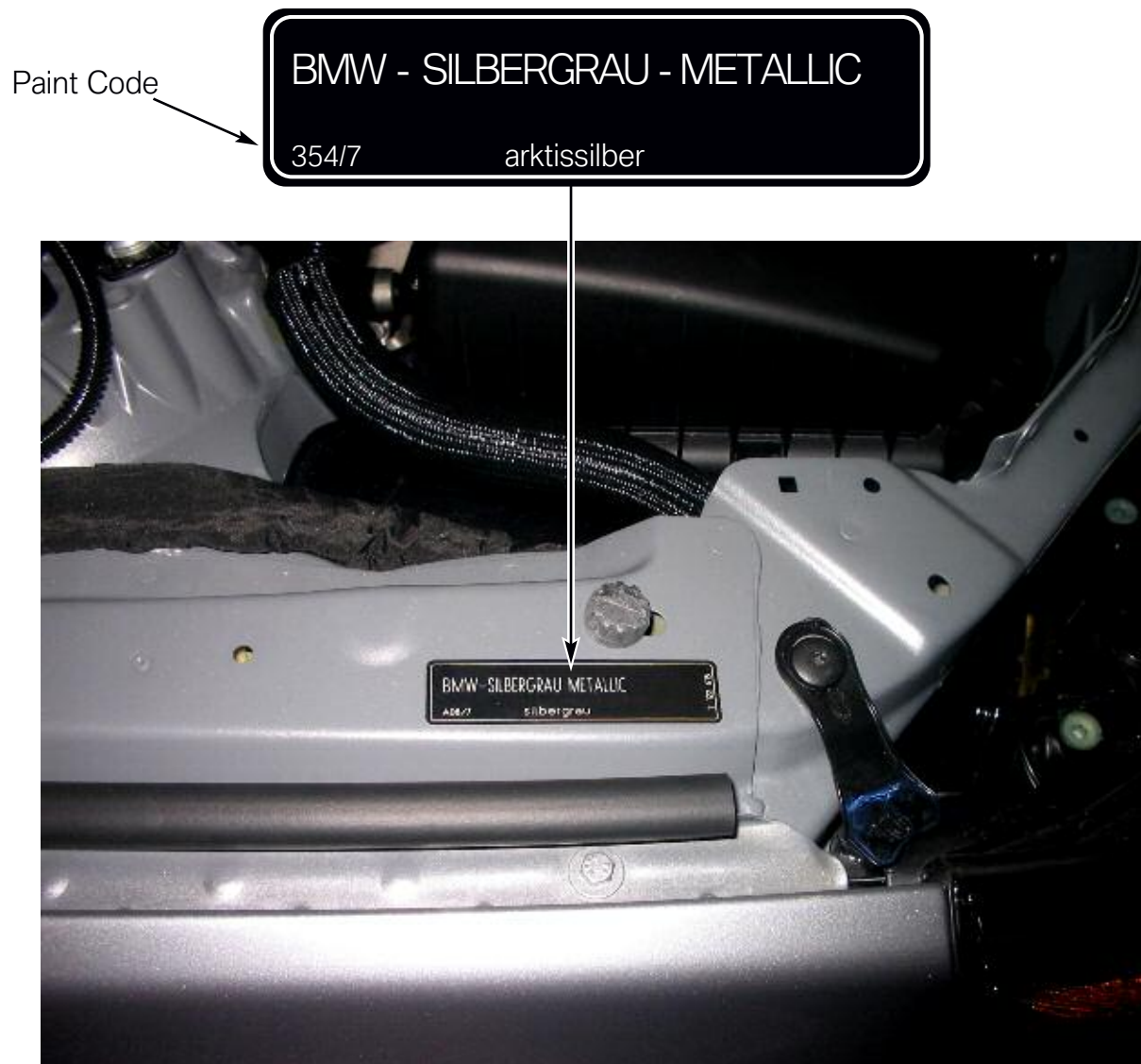
W = (Plant) Graz Austria

V = (Plant) Leipzig Germany

Digits 12-17 are the Sequential production number: ←

Paint Code Identification

When ordering painted parts it is necessary to obtain the vehicle's paint code. The paint code consists of a three digit number printed on the color identification tag as shown below. All color identification tags were located in the engine compartment up to and through out certain MY (Model Year) 2011 vehicles.



Starting with MY 2012, no vehicle had the color identification tag under the hood/engine compartment. The 3 digit paint code identification was moved to the driver side B-pillar area. It is located on the right/bottom corner of the V.I.N. sticker.



Paint Code



Engine Identification

BMW engines are identified by a combined alpha-numeric code. Like the internal model designation it is a code used by the Engineering and Technical development teams during the design stages of the engine:

- M - for standard production engines
- S - for Motorsport engines (Motorsport is a separate division of BMW)
- N, B- for New Generation engines

After the letter there are two digits that identify the engine. The table below lists engines by number of cylinders. Please refer to the BMW Technical Information Card for individual model application.

4 Cylinder	M10: 1.8 & 2.0L 2 valve inline	M42: 1.8L 4 valve inline	M44: 1.9L 4 valve inline	S14: 2.3L 4 valve inline	N20 ^{1, 2, 3} : 2.0L 4 valve inline twin-scroll single turbo	N26 ^{1, 2, 3} : 2.0L 4 valve inline twin-scroll single turbo	N47 ³ : 2.0L 4 valve inline diesel mono-scroll single VNT turbo	B46 ^{1, 2, 3} : 2.0L 4 valve inline twin-scroll single turbo	B46TU ^{1, 2, 3} : 2.0L 4 valve inline twin-scroll single turbo
	B48TU ^{1, 2, 3} : 2.0L 4 valve inline twin-scroll single turbo	6 Cylinder		M20: 2.5 & 2.7L 2 valve inline	M21: 2.4L 2 valve inline diesel turbo	M30: 2.8, 3.0, 3.2 3.4 & 3.5L 2 valve inline	M50: 2.5L 4 valve inline	M50TU: 2.5L 4 valve inline	M52 ¹ : 2.5 & 2.8 L 4 valve inline
6 Cylinder	M52TU ¹ : 2.5 & 2.8L 4 valve inline	M54 ¹ : 2.5 & 3.0L 4 valve inline	M57D30T ²¹ : 3.0L 4 valve inline diesel bi-turbo	N52(K) ¹ : 3.0L 4 valve inline	N54 ^{1 3} : 3.0L 4 valve inline bi-turbo	N55 ^{1, 2, 3} : 3.0L 4 valve inline twin-scroll single turbo	B58 ^{1, 2, 3} : 3.0L 4 valve inline twin-scroll single turbo	N57 ³ : 3.0L 4 valve inline diesel mono- scroll VNT single turbo	
	B58TU ^{1, 2, 3} : 3.0L 4 valve inline twin-scroll single turbo	B57 ³ : 3.0L 4 valve inline diesel mono- scroll VNT single turbo	S38: 3.5 & 3.6L 4 valve inline	S50: 3.0L 4 valve inline	S52 ¹ : 3.2L 4 valve inline	S54: 3.2L 4 valve inline	S55 ^{1, 2, 3} : 3.0L 4 valve inline mono-scroll bi turbo		

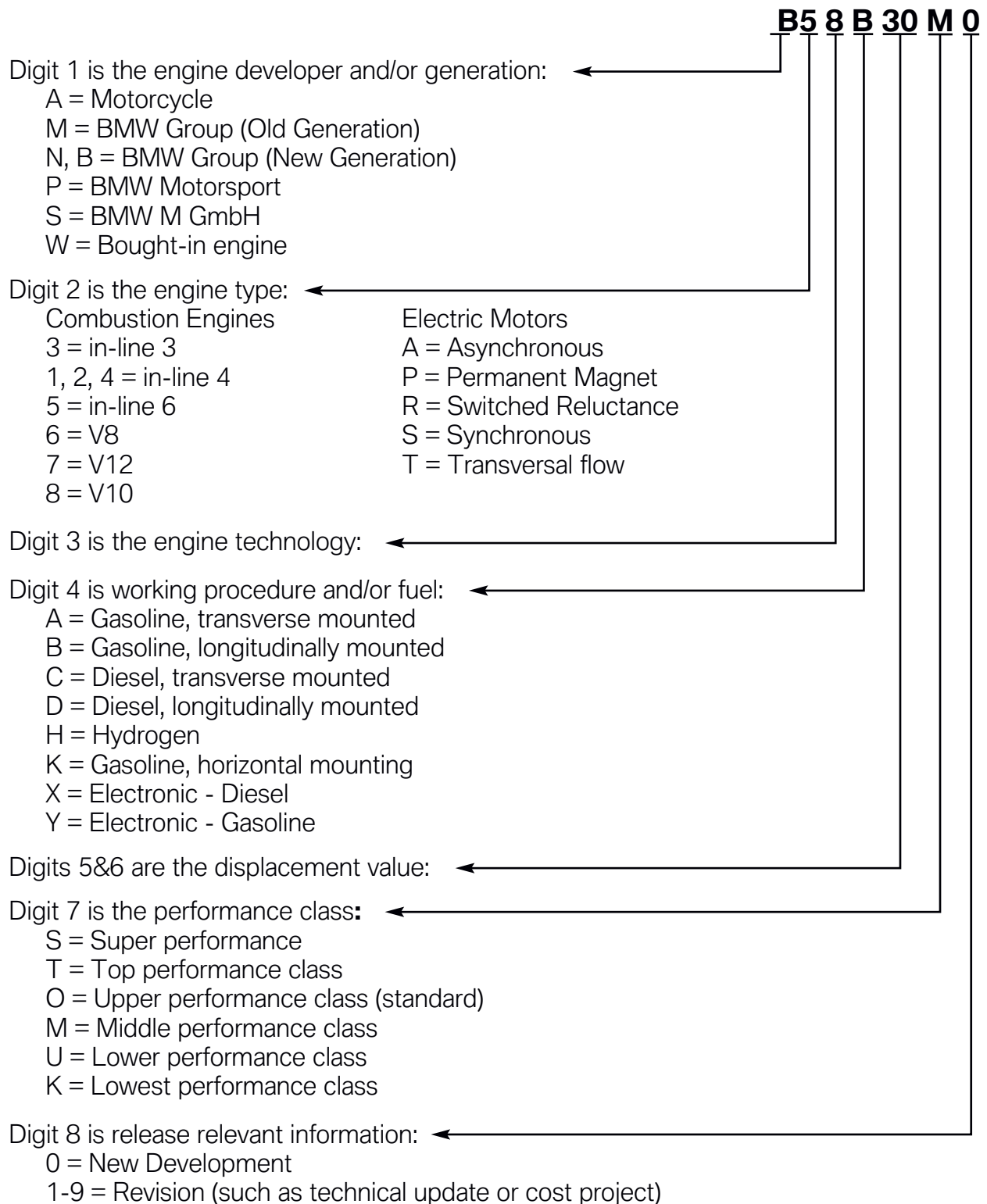
1 - VANOS: Variable Valve Timing **2** - Valvetronic: Variable Valve Timing and Lift **3** - Direct Injection

8 Cylinder	M60: 3.0 & 4.0L 4 valve V config.	M62: 4.4L 4 valve V config.	M62TU ¹ : 4.4 & 4.6L 4 valve V config.	N62 ^{1, 2} : 4.4L 4 valve V config.	N62 ^{1, 2} : 4.8L 4 valve V config.	N62TU ^{1, 2} : 4.8L 4 valve V config.	N63 ^{1, 3} : 4.4L 4 valve V config. bi-turbo reverse flow	N63TU2 ^{1, 3} : 4.4L 4 valve V config. bi-turbo reverse flow	
	N63TU3 ^{1, 3} : 4.4L 4 valve V config. bi-turbo reverse flow		S62 ¹ : 5.0L 4 valve V config.	S63 ^{1, 3} : 4.4L 4 valve V config. twin-scroll bi-turbo reverse flow	S65 ¹ : 4.0L 4 valve V config.	S63TU ^{1, 2, 3} : 4.4L 4 valve V config. twin-scroll bi-turbo reverse flow	S63TU2 ^{1, 2, 3} : 4.4L 4 valve V config. twin-scroll bi-turbo reverse flow		
10 Cylinder	S85 ¹ : 5.0L 4 valve V config.		12 Cylinder	M70: 5.0L 2 valve V config.	M73: 5.4L 2 valve V config.	M73TU: 5.4L 2 valve V config.	N73 ^{1, 2, 3} : 6.0L 4 valve V config.	N74 ^{1, 3} : 6.0L 4 valve V con- fig. bi-turbo	S70: 5.6L 2 valve V config.
3 Cylinder (i8)	B38 Top ^{1, 2, 3} : 1.5L 4 valve inline mono- scroll single turbo		2 Cylinder (i3 REX)	W20: 0.6L 4 valve inline					

1 - VANOS: Variable Valve Timing **2** - Valvetronic: Variable Valve Timing and Lift **3** - Direct Injection

Engine Numbering System

In order to organize and classify the engines developed by BMW, a numbering system has been developed.



Engine Displacement Identification

In technical literature usually only the group of 3 digits that follow the basic engine code is utilized for engine identification. These digits identify the required fuel type of the engine and its displacement.

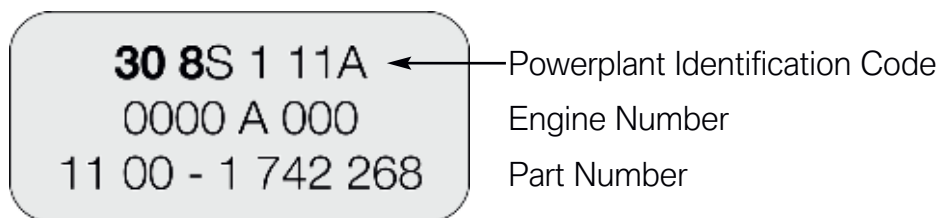
ENGINE CODE

M54-B30

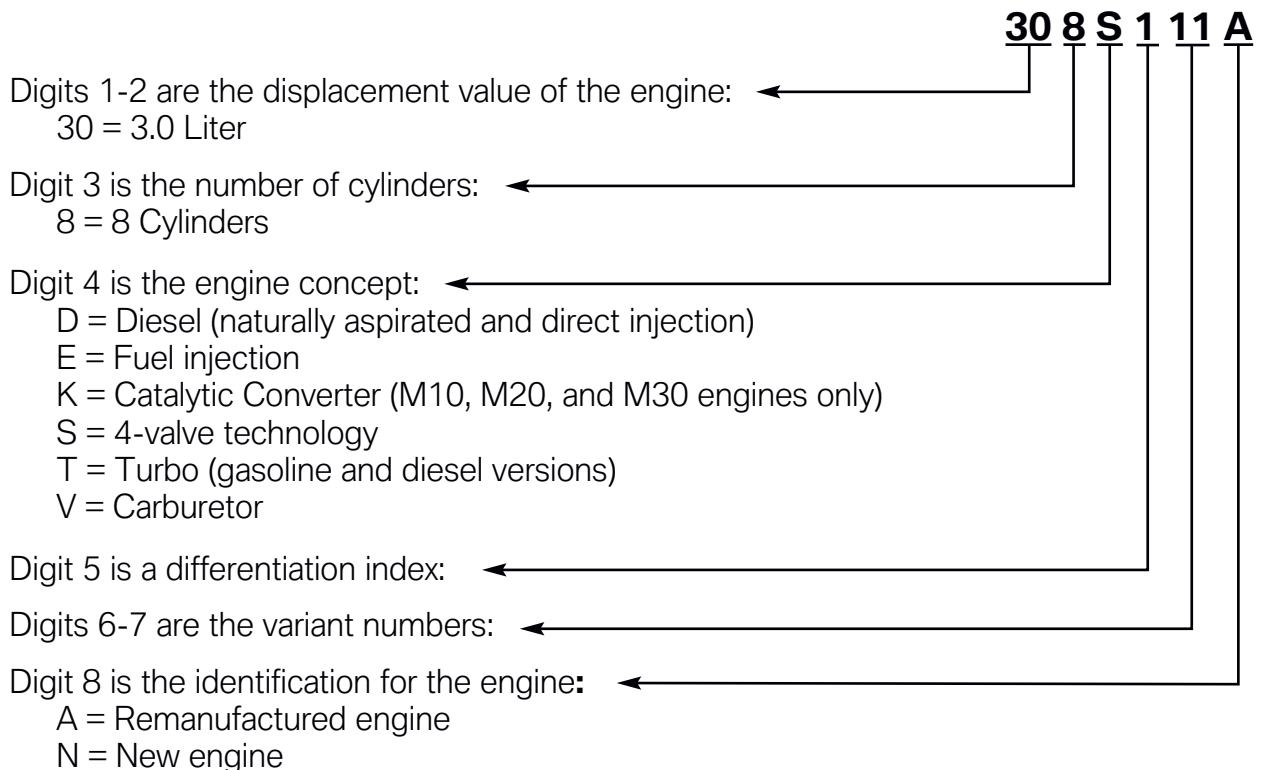
30 = 3.0 LITER
DISPLACEMENT

B = BENZINE, GERMAN
FOR GASOLINE

Engine displacement can always be determined by the engine specification plate. This plate is stamped onto the engines in various locations. There is a lot of information included in this identification plate. The breakdown of the information on this plate is as follows:



Powerplant Identification Number

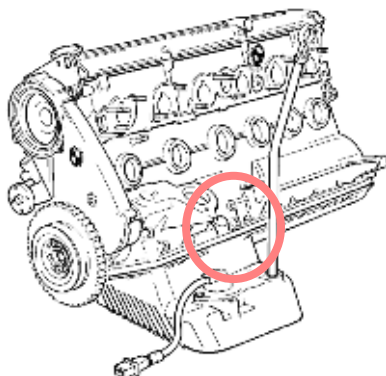


In recent time the Powerplant Identification Code changed from 8 digit to 7 digit. This mark is located on the crankcase to ensure unambiguous identification and classification. This engine identification is also necessary for approval by government authorities. Decisive here are the first seven positions. The first 6 positions are the same as the engine designation.

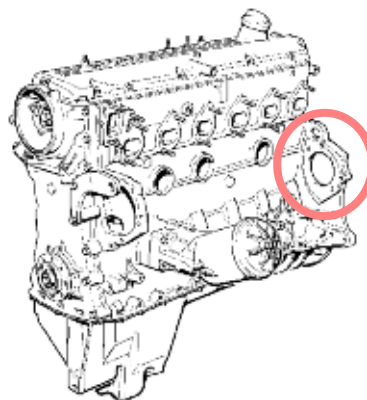
Item	Meaning	Index / explanation
1	Engine developer	M, N, B = BMW Group P = BMW Motorsport S = BMW M GmbH W = non-BMW engines
2	Engine type	5 = R6 (e.g. N55) 6 = V8 (e.g. N63) 7 = V12 (e.g. N73) 8 = V10 (e.g. S85)
3	Change to the basic engine concept	0 = basic engine 1 – 9 = changes, e.g. combustion process
4	Working method or fuel type and possibly installation position	B = petrol, longitudinal installation D = diesel, longitudinal installation H = hydrogen
5	Displacement in liters	1 = 1 liter (whole number of liters)
6	Displacement in 1/10 liter	8 = 0.8 liter (tenth of liter)
7	Type test concerns (changes that require a new type test)	A = Standard B – Z = Depending on requirement, e.g. RON 87

Engine Specification Plate Locations

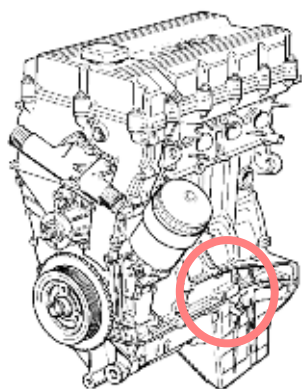
M20



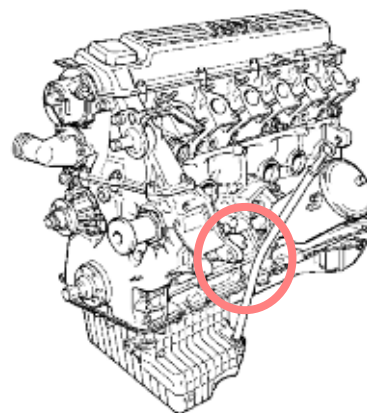
M30



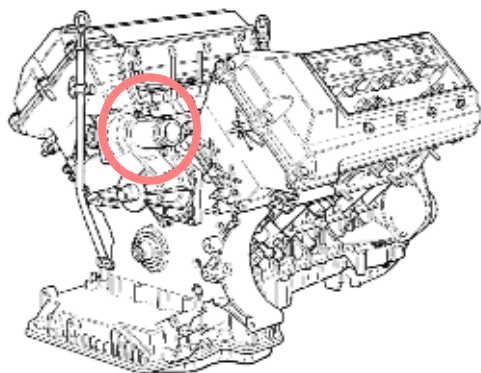
**M42
M44**



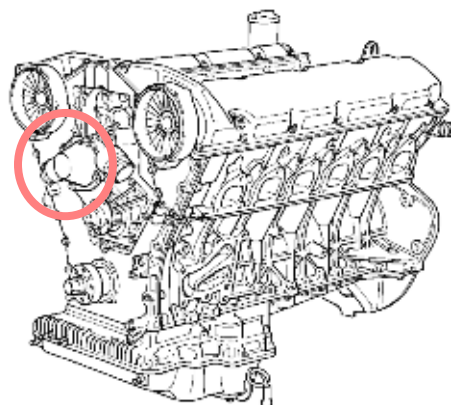
**M50
M52
M54
S50
S52
S54**



**M60
M62
N62
S62**

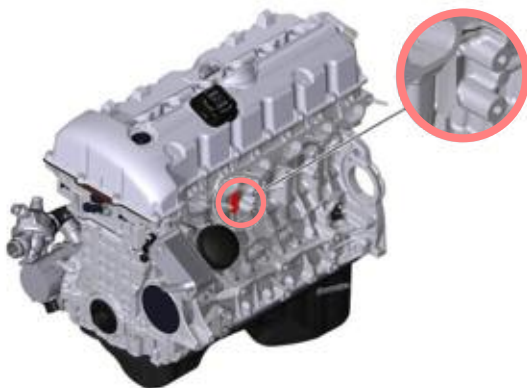


**M70
M73
S70**

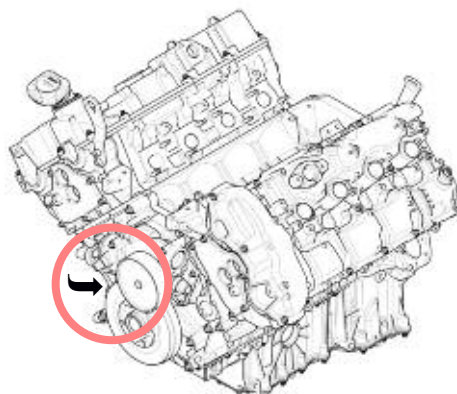


Engine Specification Plate Locations (Cont.)

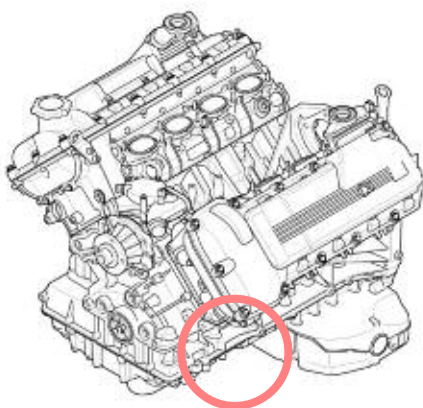
N52
N54
N55
S55



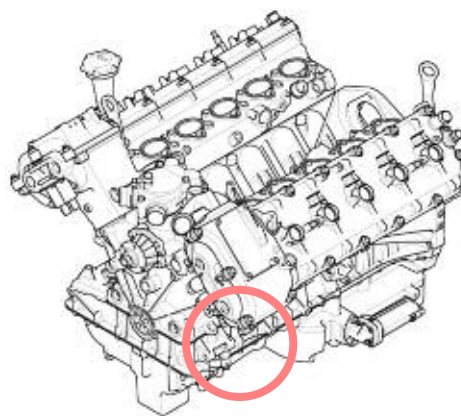
N63
S63



S65



S85



N74



N20
N26



Engine Specification Plate Locations (Cont.)

N47



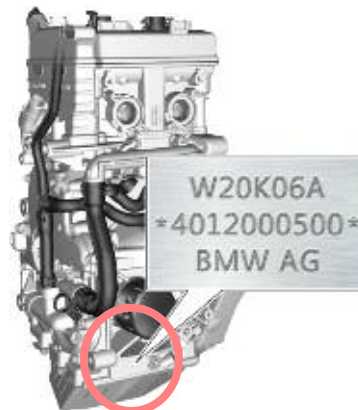
N57



B38



W20



B58

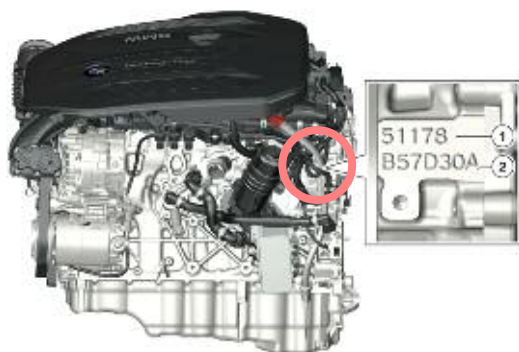


**B46
&
B48**



Engine Specification Plate Locations (Cont.)

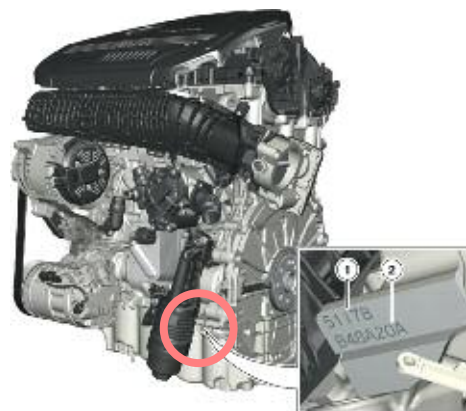
B57



B46TU

&

B48TU



Transmission Identification

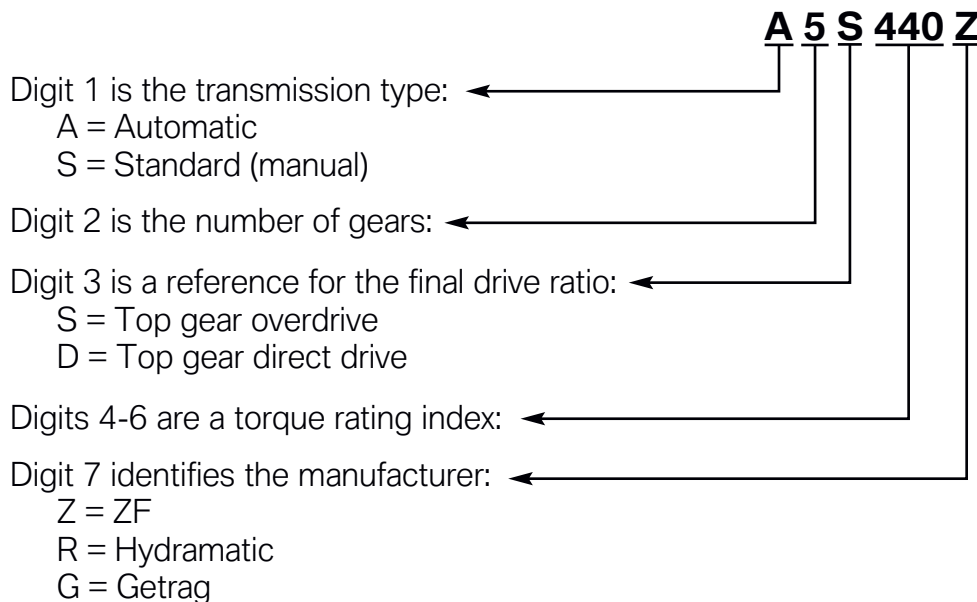
BMW uses three different transmission suppliers.

1. Zahnradfabrik Friedrichshafen: A German supplier of BMW transmissions. Commonly referred to as ZF. ZF manufactures both manual and automatic transmissions.
2. Getrag: A German supplier of BMW manual transmissions.
3. Hydramatic: A French supplier of BMW automatic transmissions. Hydramatic is a manufacturing division of General Motors Powertrain.

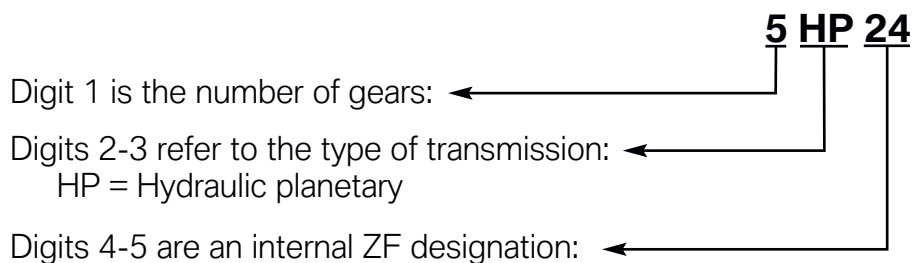
BMW has developed an internal numbering system for their transmissions for parts ordering, information research, and identification. Each manufacturer uses their own internal identification system.

The breakdown of the BMW transmission identification code is as follows:

BMW Identification Code Breakdown



ZF Identification Code Breakdown



Automatic Transmissions

In technical documentation, the transmission designation is used to ensure the unambiguous identification of the transmission. Frequently, however, only an abbreviation is used. This short form is used to assign a transmission to a transmission family. For example, we often talk about the GA8HP transmission family, which consists of several transmissions such as the GA8HP45Z, the GA8HP70Z and the GA8HP90Z.

Position	Meaning	Index	Explanation
1	Description	G	Transmission
2	Transmission type	A	Automatic Transmission
3	Number of gears	6 8	Six forward gears Eight forward gears
4-7*	Individual designations*	HP L R F21A 9 26 32 45 (ZF) 45 (GMPT) 70 90 390	Hydraulic planetary gear train Designation by GMPT Designation by BGMPT Designation of AISIN Warner 300 Nm gasoline engine 600 Nm gasoline engine 720 Nm gasoline engine 450 Nm gasoline engine, 500 diesel engine 350 Nm gasoline engine 700 Nm gasoline engine and diesel engine 900 Nm gasoline engine 390 Nm 4th gear 410 Nm gasoline engine
8	Manufacturer	A G J K R Z H	AISIN Getrag Jatco GKN GMPT ZF In-house part

* Numbers 4-7 serve for individual designation. A transmission variant, size, transferable torque and technical update can be represented here.

The tables on following three pages contain a list of automatic transmissions and the models they are used in.

Zahnradfabrik Friedrichshafen (ZF)			
Transmission BMW ID #	Manufacturer ID	Model/Year	Engine
4HP22 (EH)	4HP22 (EH)	86 535i 86 635i 86 735i	M30 M30 M30
4HP24 (EH)	4HP24 (EH)	88-9/91 750iL (E32) 92-94 750iL (E32) 90-94 850i/Ci (E31)	M70 M70 M70
A5S310Z	5HP18	93 530i/iT (E34) 94 530i/iT (E34) 95 M3 (E36) 96-99 M3 (E36)	M60 M60 S50 (US) S52
A5S325Z	5HP19	00 323i/Ci/CiC (3/00-8/00) 01 323iT (from 4/01) 01-04 325i/Ci/CiC to end of 04 02-05 325i SULEV (E46) 02- 06 325Ci SULEV (E46) 00-04 330i/Ci/CiC to end of 04 01- 03 525i/xi/xiT 3/01 to 9/03 01- 03 530i/xi/xiT 3/01 to 3/03 03-05 Z4 (E85) 2.5i and 3.0i	M52 TU M52 TU M54 M56 M56 M54 M54-N52 M54-N52 M54
A5S440Z	5HP24	97 840Ci (E31) from 9/96 97 540i (E39) 1/97-8/97 97 740i/iL (1/97 - 4/97) 97 740i/iL (5/97-8/97) 98-03 540i 98-01 740i/iL 00- X5 4.4i (E53)	M62 M62 M62 M62 M62, M62 TU 99-02 M62, M62 TU 99-01 M62TU
A5S560Z	5HP30	93-94 740i/iL (E32) 93 540i (E34) 94-95 540i (E34) 94-95 840Ci (E31) 95 740i/iL (E38) 95-01 750iL (E38) 96-97 740i/iL (- 1/97) 96 840Ci (E31) 95-97 850Ci	M60 M60 M60 M60 M60 M73/M73TU M62 M62 M73

Zahnradfabrik Friedrichshafen (ZF)

Transmission BMW ID #	Manufacturer ID	Model/Year	Engine
GA6HP26Z	6HP26	02-04 745i/Li (E65/E66) 05-08 750i/Li (E65/E66) 03- 760Li (E65/E66) 03-3/07 545i/550i (E60) 03-9/06 X5 4.4/4.8iS (E53)	N62 N62/N62TU N73 N62/N62TU N62/N62S
GA6HP28Z	6HP26TU	3/07 545i/550i (E60) 9/07 645Ci/CiC (E63/E64) 9/07 650Ci/CiC (E63/E64) 10/06 X5 4.8i (E70) 9/08 X5 xDrive35d (E70) 1/08 X6 XDrive50i 4.4 (E71)	N62/N62/TU N62/N62/TU N62/N62/TU N62TU N57 N63
GA6HP19Z	6HP19	05-9/06 325i/xi (E90/E91) 05-9/06 330i/xi (E90) 06 Z4 3.0i/si (E85) 06 Z4 3.0i/si Coupe (E86)	N52 N52 N52 N52
GA6HP21Z	6HP19TU	4/06 335i/xi (E90) 9/06 328i/xi-335i/xi (E92) 12/06 335i (E93) 3/07 528i/xi-535i/xi (E60/E61) 1/08 135is Coupe (E82) 1/08 135is Convertible (E88) 10/06 X5 3.0si (E70) 1/08 X6 XDrive35i 3.0 (E71)	N54 N52K/N54 N54 N52K/N54 N54 N54 N52K N54
GA8HP45Z	8HP45	3/10 535i/xi (F07) 3/10 528i (F10) 3/10 535i/xi (F10) 5/10 X5 xDrive35i (E70) 5/10 X5 xDrive50i (E70) 5/10 X5 xDrive35d (E70) 2/12 328xi (F30) 2/12 335xi (F30) 10/12 X5 (F15)	N55 N52 N55 N55 N63 M57 N20 N55 N55

Zahnradfabrik Friedrichshafen (ZF)				
Transmission BMW ID #	Manufacturer ID	Model/Year		Engine
GA8HP70Z	8HP70	9/09	550i/x (F07)	N63
		9/09	760Li (F02)	N74
		3/10	550i/xi (F10)	N63
		10/13	X5 (F15)	N63, N57
GA8HP75Z	8HP75	8/14	X5 (F15)	N63, N57
		8/14	X6 (F16)	N63, N57
		8/15	750i/xi (G12)	N63TU2
	M8HP75	7/14	X5 M (F85)	S63TU
		7/14	X6 M (F86)	S63TU
		11/17	M5 (F90)	S63TU2
GA8HP50Z	8HP50	8/15	740i/xii (G12)	B58
		8/17	X3 (G01)	B46
		7/17	640i GT (G32)	B58
GA8HP50Z Sport	8HP50 Sport	8/17	X3 (G01)	B58
		4/18	X4 (G02)	B46, B58

AISIN Warner (AW)				
Transmission BMW ID #	Manufacturer ID	Model/Year		Engine
GA8F22AW	AWF8F35	7/15	X1 (F48)	B46
		11/17	X2 (F39)	B46

GM Hydrumatic			
Transmission BMW ID #	Manufacturer ID	Model/Year	Engine
A4S310R (THM-R1)	4L30-E(A4S310R)	90-92 525i (E34) 93-95 525i (E34) 92 325i, is, ic (E36) 93-95 325i, is, ic (E36) 92-95 318ti (E36)	M50 M50 TU M50 M50 TU M42
A4S270R	4L30-E(A4S270R)	96-98 328i (is, ic-97) 96-98 318i (is, ic-97) 96-98 318ti (E36/5) 96-98 Z3 1.9 (E36/7) 96-98 Z3 2.8 97-98 528i (E39)	M52 M44 M44 M44 M52 M52
A5S360R (GM5)	5L40-E (A5S360R)	99-00 323i/Ci (7/98-3/00) 99-00 328i/Ci (6/98-5/00) 99-00 528i (E39) 9/99-8/00 99-00 Z3 (E36/7) 2.3/2.8	M52 TU M52 TU M52 TU M52 TU
A5S390R (GM5)	5L40-E (A5S390R)	00-8/06 X5 3.0i (4/00) 01 325iT (8/00-3/01) 01-3/05 325xi/xiT & 330Xi (-8/00) 01-6/06 325Ci/CiC/330Ci/CiC 01 525i/iT (9/00-3/01) 01 530i (9/00-3/01) 01-02 Z3 2.5/3.0 (6/00) 03-8/06 X3 2.5/3.0 (E83)	M54 M54 M54 M54 M54 M54 M54 M54
6L46R (GM6)	6 Speed Hydra-Matic	9/06 328i/xi/C/Cxi (E90/91/92) 12/06 328CiC (E93) 1/08 128is Coupe (E82) 1/08 128is Convertible (E88) 9/06 X3 3.0si (E83)	N52K/N51 N52K/N51 N52K/N51 N52K/N51 N52K

Manual Transmissions

In the technical documentation, the transmission designation is used to ensure the unambiguous identification of the transmission. Frequently, however, only an abbreviation is used. Thus we frequently speak of the I transmission, K transmission or G transmission. For the correct designation, refer to the following table.

Position	Meaning	Index	Explanation
1	Description	G	Transmission
2	Transmission type	S	Manual Transmission
3	Number of gears	1-9	Number of forward gears
4	Transmission type	- X S W D Y	Manual transmission Four-wheel drive with manual transmission Sequential manual transmission Four-wheel drive with sequential manual transmission Twin-clutch gearbox Four-wheel drive with twin-clutch gearbox
5 & 6	Transmission type	17 26 37 45 53	I transmission D transmission H transmission K transmission G transmission
7	Gear set	B D S P	Petrol engine ratio Diesel engine ratio Sport gear ratio Petrol engine gear ratio overhauled
8	Manufacturer	G J R Z H	Getrag Jatco GMPT ZF In-house part

The following table lists some of the manual transmissions and the vehicles they are used in.

Transmission Model	Vehicles	Engine	Model Year
S5D 200G	318i/iC (E36)	M42	1992 - 1995
S5D 250G	318ti (E36/5) 318i/iC (E36) 325i/iC (E36) 323i (E46)	M42, M44 M44 M50TU M52TU	1995 - 1999 1996 - 1992 - 1995 1999 -
S5D 280Z	X5 3.0i (E53) 325xi (E46/16) 330xi (E46/16)	M54 M54 M54	2001 - 2001 - 2001 -
S5D 310Z	M3 (E36) 525i (E34) 530i (E34)	S50 M50TU M60	1995 - 1991 - 1995 1994 - 1995
S5D 320Z	323i/iC, 328i/iC (E36) M3 (E36) Z3 roadster/Coupe M roadster/Coupe 528i (E39) 328i (E46)	M52 S52 M52TU S52 M52TU M52TU	1996 - 1999 1996 - 1999 1997 - 1998 - 1997 - 1999 -
S6S 420G	M3 (E46) 540i (E34) 540i (E39) M5 (E39) Z8 (E52)	S54 M60 M62TU S62 S62	2001 - 1994 - 1995 1997 - 2000 - 2001 -
S6S 560G	850i/Ci 850Csi	M70, M73 S70	1991 - 1997 1994 - 1995
GS6-37BZ	330i (E46) 525i (E60) 328i (E90)	M54 M54 N52	2001 2003 2006 -
GS6-17PG	528i (F10)	N52	2011 -
GS6-45BZ	535i (F10) 335i (F30)	N55 N55	2011 - 2012 -
GS6-53BZ	M3 (E9x) 550i (F10)	S65 N63	2008 - 2011 -

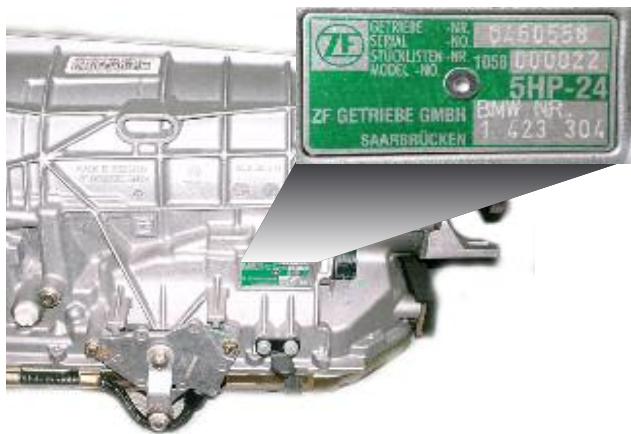
Transmission Model	Vehicles	Engine	Model Year
GS6-17BG	328i (F30)	N20	2012 -
GS7-D36SG	M3 (E9x)	S65	2008 -
GS7-D36BG	M5 (F10) M6 (F12/13) M3 (F8x)	S63TU S63TU S55	2013 2013 2015 -

Transmission Data Plate

In order to identify BMW transmissions there are identification tags located externally on the transmission case. The tag contains information such as Manufacturer, Serial number, transmission type etc. This information is needed when ordering parts, referencing bulletins, or submitting a PuMA case.

ZF Tag Location

- Right hand side (passenger side) of transmission case. (5HP30 and 5HP18)
- Left hand side (drivers side) of transmission case. (5HP26, 6HP19, 6HP45, 8HP7x)
- Rear under output shaft. (5HP19)



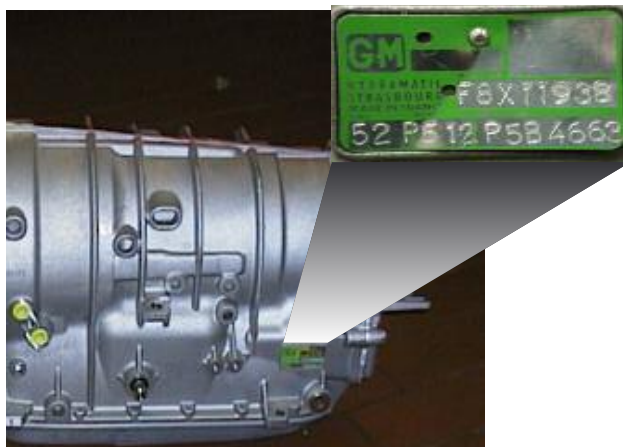
ID Tag Location 5HP24



ID Tag Location 5HP19

GM Tag Location

- Left hand side (drivers side) of transmission case.



GM ID Tag (GM 5)

BMW Group Numbering System

BMW uses a group numbering system to aid internal and BMW Center personnel to identify vehicle components or BMW information resources.

The main group numbering system is made of two digits at the beginning of any BMW group number. The following information resources are all organized by the group numbering system.

- Part Numbers
- Service Information Bulletins
- Repair Manuals
- Tightening Torques
- Electrical Troubleshooting Manuals
- Warranty Labor and Defect codes
- Technical Reference Information
- Technical Data
- Special Tool Catalog
- Diagnosis Program Test Modules

Group	Description	Group	Description
0	Maintenance and General Information	33	Rear Axle
1	Warranties	34	Brakes
2	Aftersales Development	35	Pedals
3	Technical Training	36	Wheels and Tires
4	Tools and Equipment	37	Special Suspension Systems
5	Information Systems	41	Body
6	Service Roundtable	51	Body Equipment
7	Diagnosis	52	Seats
8	Recall Management	54	Special Roofs
9	Programming/Coding	61	General Electrical Systems
10	TIS	62	Instruments
11	Engine	63	Lights
12	Engine Electrical Systems	64	Heating & Air Conditioning
13	Fuel Systems	65	Audio, Navigation, Information, Safety Systems
16	Fuel Supply Systems	66	Driver Assistance, Remote Control Systems
17	Cooling System	67	Electric drives
18	Exhaust Systems	71	Tools and Accessories (Engine-Chassis)
21	Clutch	72	Safety Belts and Accessories (Body)
22	Engine and gearbox suspension	80	Bicycles/Bicycle Accessories
23	Manual Transmission	81	Books, Calendars and Collectibles
24	Automatic Transmission	82	Lifestyle and Gift Items
25	Gear Shift Mechanism	84	Communication Systems
26	Drive Shaft	91	Individual Equipment
27	Intermediate and Special Transmission	97	Corrosion Protection
28	M Double-clutch Transmission w/Drivelogic	98	Paints and Car Care Products
31	Front Axle	99	Paint Work
32	Steering and Wheel Alignment		

Basic BMW Abbreviations and Acronyms

Throughout your career as a BMW Group Service Technician you will come across many acronyms in the various Technical publications. Even BMW is an acronym. An acronym is a word formed from the initial letters of a longer word. In the next two pages there is a list of acronyms describing various systems on BMW Group automobiles and SAVs.

Remember this is only a partial list. The BMW Group is a dynamic company and is always adding new systems to its products. Look for new systems and acronyms with new Technical Training courses, Service Information, and Technical Information Bulletins. **You can also refer to ISTA “FTD 000 103035 Abbreviations All models” for more than 1500 definitions.**

AB	Airbag	CVM	Convertible Top Module
ABL	Procedure	CVT	Constantly Variable Trans.
ABS	Anti-lock Braking System		
ACSM	Advanced Crash Safety Module	D-Bus	Diagnosis Bus (same as TXD)
ADS	Engine Intake Air Control	D-CAN	Diagnosis on Controller Area Network
AGS	Adaptive Transmission Control	DBC	Dynamic Brake Control
AIC	Automatic Interval Control (rain sensor)	DCS	Dealer Communication System
AIR	Aftersales Information Research	DDE	Digital Diesel Electronics
AL	Active Steering (also AS or AFS)	DIN	German Industrial Standards
ARS	Active Roll Stabilization	DIS	Diagnosis and Information System
ASC	All Season Traction	DISA	Differential Air Intake Control
ASC+T	ASC+Traction	DK	Throttle Housing
ASK	Audio System Controller	DKI	Throttle Position
AUC	Automatic Air Recirculation	DME	Digital Motor Electronics
AZD	Tightening Torque (AZD)	DM-TL	Diagnosis Module Tank Leakage
		DSC	Dynamic Stability Control
B	Benzene (gasoline)	DSP	Digital Sound Processing
BC	Board Computer	DTC	Diagnostic Trouble Code
BC1	Body Controller 1	DWA	Theft Deterrent System
BDC	Body Domain Controller		
BLS	Brake Light Switch	EBO	Installation Location
BMBT	Board Monitor	EBV	Electronic Brake Proportioning
BS	Block Diagram	ECM	Engine control module (SAE)
BSD	Bit Serial Data interface	EDC	Electronic Dampening Control
BST	Battery Safety Terminal	EDK	Electronic Throttle Valve
BZM	Center Console Control Center	EGS	Electronic Transmission Control
		EH	Electronic Hydraulic
CAS	Car Activation System	EHC	Electronic Height Control
CBC	Corner Braking Control	EKP	Electronic Fuel Pump
CAN-Bus	Controller Area Network(bus)	EKPS	Electronic Fuel Pump Control
CIC	Car Information Computer	ELV	Electronic Steering Lock
CID	Central Information Display	EM	Electro-Mechanical
CIM	Chassis Integration Module	EMF	Electro-Mechanical Emergency Brake

EML	Electronic Motor Load Regulation	JBE	Junction Box Electronic
EO	Component Location		
EPC	Electronic Parts Catalog	KAFAS	Camera-based driver assistance system
ETM	Electronic Troubleshooting Manual	KAT	Catalytic Converter
Ethernet	Cabled data network technology for local data networks	KATON	Converter Heating (signal)
EWS	Electronic Driveway Protection	K-Bus	Body Bus
		K-CAN	Body controller area network
FB	Function Description	KFA	Function changes to vehicle
F-CAN	Chassis CAN	KL	Terminal Designation
FD	Rear display	KO	Compressor "ON" Signal
FEB	Fault Elimination	KOMBI	Instrument Cluster
FEM	Front Electronic Module	KOREL	Compressor Relay Signal
FBZV	Radio Frequency Locking	KW	Crankshaft
FLA	High beam assistant		
FlexRay	Fast, present and fault-tolerant bus system for use in automotive applications	LCM	Lamp Check Module
FRU	Flat Rate Unit	LDP	Leak Diagnosis Pump
FRM	Footwell module	LEV	Low Emissions Vehicle
FTD	Vehicle Technical Diagnosis	LIN-Bus	Local Interconnect Network bus
FUB	Functional Description	Local-CAN	Local controller Area Network
FZD	Roof function center	LL	Closed Throttle
		LRA	Vertical Headlight Aiming
GAL	Speed Dependent Volume	LSM	Steering Column Memory
GM	General Module	LSZ	Lamp Switching Center
GMR	Yaw Moment Control	LVA	Air Supply System
GPS	Global Positioning System	LWR	Vertical Headlight Aiming
GWS	Gear Selector Switch	LWS 5	Steering Angle Sensor
HDC	Hill Decent Control	MBC	Maximum Brake Control
HFM	Hot Film Air Mass Meter	MFL	Multi-Function Steering Wheel
HKL	Luggage compartment lid lift	MFC	Multi-Function Controller
HSB3	Heating/Ventilation, 3rd row	MID	Multi-Information Display
HSR	Rear suspension slip angle control	MOST	Media Oriented System Transport
HUD	Head Up Display	MRS	Multiple Restraint System
		MSR	Engine Drag Torque Reduction
I-Level	Integration level	NG	New Generation
IB	Interior lighting Control Signal	NVE	Night Vision Electronics
ICM	Integrated Chassis Management		
IHKA	Automatic Heating and A/C	OBC	On-Board Computer
IHKR	Regulated Heating and A/C	OBD	On-Board Diagnosis (SAE)
IHKS	Standard Heating and A/C		
IKE	Instrument Cluster Electronics	PB	Pin Assignments
IR	Infrared	P-Bus	Periphery Bus
ISN	Individual Serial Number	PDC	Park Distance Control
ISTA	Intergrated Service Technical Application	PIB	Pin Assignment
		PT-CAN	Powertrain CAN
		PWG	Pedal Position Sensor

RDC Tire Pressure Control
 RDW Tire Pressure Warning
 REP Repair Instructions
 RPS Rollover Protection System
 RPW Tire Puncture Warning
 RSE Rear Seat Entertainment system
 RXD Wake-up Diagnosis Line
 RZV Direct Stationary Ignition

SB Fuse Assignments
 SBE Seat Occupancy Detector
 SCA Soft Close Actuator
 SDARS Satellite tuner (US)
 SG Control Unit
 SGS Seat Integrated Belt System
 SHD Sunroof Module
 SIA Service Interval Indicator
 SIT Service Information Technology
 SM Seat Module
 SP Schematic
 SPI Vehicle Software Information
 ST Connector View
 STA Connector View
 SSP Wiring Diagram
 SWW Lane Change Warning
 SWZ Special Tool
 SZL Steering column switch cluster
 SZM Central Switch Center Module

TCU Telematic Control Unit
 TD Engine Speed
 TED Technical Data
 ti Injector on time
 TL Part Throttle
 TLEV Transitional Low Emission Vehicle
 TPMS Tire Pressure Monitoring System
 TRS Battery Isolation Switch
 TRSVC Control unit for reversing and side view cam-eras
 TU Technical Update
 TVDI Turbo Valvetronic Direct Injection
 TXD Transmitting Diagnosis Line

ULF-SBX Universal charger and hands-free unit, interface box (bluetooth telephony)

VANOS Variable Camshaft Timing
 VDM Vertical Dynamics Management
 VL Full load - wide open throttle
 VSW Video Switch
 VTG Transfer box

WK Converter Lock-up Clutch
 WUP Wake Up line

ZGM Central Gateway Module
 ZKE Central Body Electronics
 ZV Central Locking System
 ZWD Idle Control Valve

Terminals

B+ Battery Positive
 B- Battery Negative
 KL Standardized Abbreviation for Clamp or Terminal Number
 KL_0 Ignition Switch Off
 KL_R Voltage Ignition Switch in ACC, Run Start. Hot in ACC/Run/Start
 KL_15 Voltage Ignition Switch in Run and Start. Hot (12V) in Run/Start.
 KL_15N F01/F02. Voltage Ignition Switch in Run and Start. Hot (12V) in Run/Start
 KL_15U/15 Voltage Ignition Switch in Run. Hot (12V) in Run.
 KL_30 12V at All Times (Relay Work Power). Hot (12V) All Times.
 KL_30g 12V with time disconnection.
 KL_30B F01/F02. 12V with time disconnection.
 KL_30g-f 12V with fault disconnection.
 KL_30F F01/F02. 12V with fault disconnection.
 KL_30H Starter Signal
 KL_31 Ground
 KL_50 Voltage Ignition switch in Start. Hot (12V) in Start
 KL_58 Interior Lighting Dimmer Signal
 KL_61 Ground with Alternator Output, 12V with KL_15
 PAD Testing/Analysis/Diagnosis (G12+)
 85 Relay Coil Ground (Signal) control side.
 86 Relay Coil B+ control side.
 87 Relay Output Work Side.
 87a Relay Output Work Side At Rest